

Quiz-2

Q: Two customer support agents in a software company are handling service requests. Let the random variable **X** represent the number of attempts Agent 1 requires to successfully resolve a customer issue. Similarly, let the random variable **Y** denote the number of attempts Agent 2 requires to resolve their assigned issue.

For Agent 1: **X** are 1, 2, 3, 4 with probabilities 0.35, 0.30, 0.20, 0.15

For Agent 2: **Y** are 1, 2, 3 with probabilities 0.5, 0.3, 0.2

Assuming that the performances of two agents are independent.

1. Construct joint probability distribution of **X** and **Y**.
2. Find the probability that both agents collectively resolve their issue in at most 3 attempts.